

Physical Chemistry Practical: Topic 8 — Reaction Kinetics

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 — Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Investigation of Reaction Rate by Gas Evolution: Magnesium and Hydrochloric Acid

Cambridge Syllabus Reference: 9701/33/M/J/24/Q2 · [10 Marks]

Reaction: $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$

Method: 0.050 g Mg ribbon is added to 25.0 cm³ of 1.00 mol dm⁻³ HCl. The volume of hydrogen gas collected in the syringe is recorded every 10 seconds for 120 seconds.

(a) Explain why the rate of reaction is fastest at $t = 0$ s and decreases continuously over time. [3]

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(b) Sketch the expected curve of gas volume against time. [2]

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(c) State what feature of the graph indicates that the reaction has finished. [2]

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(d) Calculate the maximum theoretical volume of H₂ gas produced at room conditions. [Ar: Mg = 24.3, V_m = 24.0 dm³ mol⁻¹] [3]

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Question 2: Determination of Initial Reaction Rate via Graphical Tangent Construction**Cambridge Syllabus Reference: 9701/32/M/J/23/Q2 · [10 Marks]**

A curve of volume of gas (V / cm^3) versus time (t / s) is plotted. A tangent drawn to the curve at $t = 0 \text{ s}$ passes through coordinates $(0, 0)$ and $(25, 45)$.

(a) Calculate the initial rate of reaction from the tangent coordinates, including units. [3]

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(b) Explain why initial rate (at $t = 0$) is preferred over average rate over the entire reaction. [3]

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(c) Describe how the tangent at $t = 0 \text{ s}$ should be drawn accurately on an experimental curve. [2]

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(d) If the temperature of the acid were increased by 10°C , state how the tangent slope would change. [2]

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Question 3: The Sodium Thiosulfate and Hydrochloric Acid Disappearing Cross Reaction

Cambridge Syllabus Reference: 9701/34/O/N/23/Q2 · [10 Marks]

Reaction: $\text{Na}_2\text{S}_2\text{O}_3(\text{aq}) + 2\text{HCl}(\text{aq}) \rightarrow 2\text{NaCl}(\text{aq}) + \text{SO}_2(\text{g}) + \text{S}(\text{s}) + \text{H}_2\text{O}(\text{l})$

Method: A conical flask is placed over a paper sheet marked with a black cross (+). The time taken, t , for the precipitated sulfur to obscure the cross is measured.

(a) Explain why the quantity $1/t$ can be taken as a direct measure of the rate of this reaction. [3]

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(b) State the physical state of the sulfur that causes the cross to disappear. [2]

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(c) If the cross disappears in 42.0 seconds, calculate the rate ($1/t$) to three significant figures with units. [2]

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(d) State two subjective errors associated with human observation of the disappearing cross. [3]

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Question 4: Determination of Reaction Order with Respect to Sodium Thiosulfate**Cambridge Syllabus Reference: 9701/33/O/N/22/Q2 · [10 Marks]**

Five runs are performed using different volumes of $0.100 \text{ mol dm}^{-3}$ $\text{Na}_2\text{S}_2\text{O}_3$ made up to 50.0 cm^3 with water, then mixed with 5.0 cm^3 of 2.0 mol dm^{-3} HCl . Results:

- Vol $\text{Na}_2\text{S}_2\text{O}_3$: $50.0, 40.0, 30.0, 20.0, 10.0 \text{ cm}^3$
- $1/t$ ($\times 10^{-3} \text{ s}^{-1}$): $44.0, 35.2, 26.5, 17.6, 8.8$

(a) Plot or describe the relationship between Rate ($1/t$) and volume of $\text{Na}_2\text{S}_2\text{O}_3$. [3]

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(b) Deduce the order of reaction with respect to $\text{Na}_2\text{S}_2\text{O}_3$ and justify your conclusion mathematically. [3]

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(c) Explain why water was added to each mixture to maintain a total volume of 55.0 cm^3 . [2]

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(d) Write the rate equation with respect to $\text{Na}_2\text{S}_2\text{O}_3$ assuming it is first order. [2]

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Question 5: Investigation of the Order of Reaction with Respect to Hydrochloric Acid**Cambridge Syllabus Reference: 9701/35/M/J/24/Q2 · [10 Marks]**

The concentration of $\text{Na}_2\text{S}_2\text{O}_3$ is kept constant, while the volume of 2.0 mol dm^{-3} HCl is varied from 5.0 to 20.0 cm^3 (total volume maintained at 50.0 cm^3). The reaction time t remains virtually unchanged at $28.0 \pm 0.5 \text{ s}$ across all trials.

(a) Deduce the order of reaction with respect to HCl based on this data. [2]

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(b) Explain what 'zero order' signifies regarding the rate-determining step of the reaction mechanism. [3]

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(c) Combine the findings from Q4 and Q5 to write the overall rate equation. [2]

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(d) State the overall order of the reaction and deduce the units of the rate constant, k . [3]

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Question 6: Experimental Design: The Critical Role of Constant Depth and Volume in Turbidimetric Experiments

Cambridge Syllabus Reference: 9701/31/O/N/23/Q2 · [10 Marks]

A student repeats the disappearing cross experiment but forgets to add distilled water to make up the volume. Consequently, the total liquid volume decreases from 55 cm^3 in trial 1 to 15 cm^3 in trial 5.

(a) Explain how varying liquid depth fundamentally invalidates the $1/t$ approximation of rate. [4]

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(b) Predict how the observed reaction time in trial 5 will compare to a properly volume-compensated trial. [2]

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(c) Suggest how a datalogger with a colorimeter can replace human visual judgment to give continuous absorbance curves. [4]

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Question 7: Effect of Temperature on Reaction Rate and Qualitative Estimation of Activation Energy

Cambridge Syllabus Reference: 9701/32/O/N/21/Q2 · [10 Marks]

The reaction between sodium thiosulfate and hydrochloric acid is performed at temperatures between 20°C and 55°C.

Results: 20°C → 56 s; 30°C → 28 s; 40°C → 14 s; 50°C → 7 s.

(a) Identify the mathematical trend shown by the reaction time as temperature increases in 10°C intervals. [2]

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(b) Explain this exponential rate increase in terms of the Maxwell-Boltzmann distribution of molecular energies. [4]

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(c) Sketch the Maxwell-Boltzmann distribution curve for two temperatures, T₁ and T₂ (where T₂ > T₁), labeling E_a. [4]

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Question 8: Quantitative Analysis of Stopwatch Precision and Human Reaction Time Uncertainty

Cambridge Syllabus Reference: 9701/34/O/N/22/Q2 · [10 Marks]

A digital stopwatch displays time to 0.01 seconds. Human reaction time is approximately ± 0.2 seconds. In a kinetics run at 50°C, the recorded time was 6.20 s; in a run at 15°C, the recorded time was 120.00 s.

(a) Explain why quoting stopwatch readings to 0.01 s in practical chemistry is misleading. [2]

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(b) Calculate the percentage uncertainty in reaction time for the 6.20 s run (assume ± 0.2 s uncertainty). [3]

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(c) Calculate the percentage uncertainty for the 120.00 s run. [2]

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(d) Explain why fast reactions ($t < 10$ s) should be avoided when designing kinetics experiments. [3]

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Question 9: Chemical Safety: Hazard Mitigation of Sulfur Dioxide Gas in Reaction Kinetics

Cambridge Syllabus Reference: 9701/31/M/J/24/Q2 · [10 Marks]

The reaction of thiosulfate with acid produces toxic sulfur dioxide, SO_2 , which is an irritant to the respiratory system. The Cambridge practical instructions mandate placing flasks into a 'stop bath' immediately after each run.

(a) State the hazard classification of sulfur dioxide and identify which students are particularly at risk. [2]

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(b) Identify the active chemical present in a standard laboratory stop bath and explain how it neutralizes SO_2 . [3]

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(c) State two laboratory ventilation precautions when carrying out thiosulfate kinetics. [2]

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(d) Write the balanced chemical equation for the reaction of SO_2 with sodium hydroxide solution. [3]

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Question 10: Homogeneous Catalysis of the Peroxodisulfate–Iodide Reaction by Transition Metal Ions

Cambridge Syllabus Reference: 9701/35/M/J/23/Q2 · [10 Marks]

Uncatalyzed Reaction: $\text{S}_2\text{O}_8^{2-}(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{SO}_4^{2-}(\text{aq}) + \text{I}_2(\text{aq})$

This reaction is very slow at room temperature because both reactants are negatively charged anions that repel each other.

The reaction can be catalyzed by either $\text{Fe}^{2+}(\text{aq})$ or $\text{Fe}^{3+}(\text{aq})$ ions.

(a) Explain why the uncatalyzed reaction between peroxodisulfate and iodide ions has a high activation energy. [2]

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(b) Write the two consecutive chemical equations demonstrating how Fe^{3+} acts as a homogeneous catalyst. [4]

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(c) Explain why each step in the catalyzed mechanism has a lower activation energy than the uncatalyzed reaction. [2]

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(d) Explain why iron(II) ions, Fe^{2+} , are equally effective as a catalyst as iron(III) ions. [2]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Physical Chemistry Practical: Topic 8 — Reaction Kinetics

Question 1: Investigation of Reaction Rate by Gas Evolution: Magnesium and Hydrochloric Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	At $t = 0$, the concentration of HCl and available surface area of Mg are at their highest, leading to maximum collision frequency [1.5]; as reactants are consumed, concentration decreases, collisions per second decrease, and the rate slows down [1.5].	[3]
(b)	Smooth curve passing through origin, steep initial gradient, leveling off horizontally at constant maximum volume [2].	[2]
(c)	The curve becomes completely horizontal / gradient equals zero [2].	[2]
(d)	Moles Mg = $0.050 / 24.3 = 2.058 \times 10^{-3}$ mol [1]; Moles HCl = 0.0250 mol (excess) [1]; Vol = $2.058 \times 10^{-3} \times 24000 = 49.4 \text{ cm}^3$ [1].	[3]

Question 2: Determination of Initial Reaction Rate via Graphical Tangent Construction [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Gradient = $\Delta V / \Delta t = (45 - 0) / (25 - 0) = 1.80$ [2]; Units = $\text{cm}^3 \text{ s}^{-1}$ (or cm^3 / s) [1].	[3]
(b)	At $t = 0$, the concentrations of reactants are precisely known (identical to initial starting concentrations) before any significant consumption or back-reaction occurs [3].	[3]
(c)	Place a transparent ruler so it follows the initial curvature smoothly at the origin, ensuring angles between ruler and curve on both sides are balanced [2].	[2]
(d)	The slope of the tangent would be significantly steeper (approximately double in gradient) due to higher kinetic energy and collision frequency [2].	[2]

Question 3: The Sodium Thiosulfate and Hydrochloric Acid Disappearing Cross Reaction [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Rate is defined as change in concentration / time [1]; the cross disappears when a constant fixed mass/concentration of solid sulfur precipitate forms [1]; since $\Delta[S]$ is constant, Rate is directly proportional to $1/t$ [1].	[3]
(b)	Colloidal suspension of solid sulfur precipitate, S(s) [2].	[2]
(c)	$1 / 42.0 = 0.0238 \text{ s}^{-1}$ [2].	[2]
(d)	Observer judgment of the exact moment the cross is obscured varies between individuals [1.5]; ambient lighting changes and viewing angle affect perception [1.5].	[3]

Question 4: Determination of Reaction Order with Respect to Sodium Thiosulfate [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	A straight line passing directly through the origin (0, 0) [3].	[3]
(b)	Order is 1 (first order) [1]; Rate is directly proportional to $[\text{Na}_2\text{S}_2\text{O}_3]^1$; doubling volume from 20 to 40 cm^3 doubles rate from 17.6 to $35.2 \times 10^{-3} \text{ s}^{-1}$ [2].	[3]
(c)	To ensure the volume of $\text{Na}_2\text{S}_2\text{O}_3$ is directly proportional to its concentration and to keep the depth of liquid constant in the flask [2].	[2]
(d)	Rate = $k [\text{Na}_2\text{S}_2\text{O}_3]$ [2].	[2]

Question 5: Investigation of the Order of Reaction with Respect to Hydrochloric Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Zero order (order = 0) with respect to HCl [2].	[2]

(b)	H ⁺ ions are not involved in the slow rate-determining step of the reaction mechanism (or are involved in a fast step subsequent to the rate-determining step) [3].	[3]
(c)	Rate = k [Na ₂ S ₂ O ₃] [2].	[2]
(d)	Overall order = 1 + 0 = 1 [1]; Units of k = (mol dm ⁻³ s ⁻¹) / (mol dm ⁻³) = s ⁻¹ [2].	[3]

Question 6: Experimental Design: The Critical Role of Constant Depth and Volume in Turbidimetric Experiments [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	The black cross is obscured when the total mass of sulfur in the light path reaches a threshold [1]; if depth decreases, light travels through less liquid [1]; more sulfur precipitate per unit volume is required to obscure the cross, so $\Delta[S]$ is no longer constant [2].	[4]
(b)	The time will be significantly longer than expected [2].	[2]
(c)	Light from an LED source passes through a cuvette containing the mixture to a photodiode detector [2]; light transmittance decreases smoothly as sulfur precipitate scatters light, allowing the computer to plot absorbance against time continuously [2].	[4]

Question 7: Effect of Temperature on Reaction Rate and Qualitative Estimation of Activation Energy [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	For every 10°C rise in temperature, the reaction time halves (rate doubles) [2].	[2]
(b)	At higher temperature, average kinetic energy increases and the distribution curve flattens and shifts to the right [1]; a significantly greater fraction of molecules possess kinetic energy greater than or equal to the activation energy ($E \geq E_a$) [2]; the frequency of successful collisions per unit time increases dramatically [1].	[4]
(c)	Curves start at origin, peak of T ₂ lower and shifted right [2]; E_a marked on energy axis, shaded area for T ₂ noticeably larger than for T ₁ [2].	[4]

Question 8: Quantitative Analysis of Stopwatch Precision and Human Reaction Time Uncertainty [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Human reaction time in pressing start and stop buttons is around ± 0.2 s, which is 20 times larger than the stopwatch display resolution [2].	[2]
(b)	% uncertainty = $(0.2 / 6.20) \times 100 = 3.23\%$ [3].	[3]
(c)	% uncertainty = $(0.2 / 120.0) \times 100 = 0.17\%$ [2].	[2]
(d)	In fast reactions, percentage human timing uncertainty becomes very large ($>5\%$), and mixing time becomes a significant fraction of total reaction time [3].	[3]

Question 9: Chemical Safety: Hazard Mitigation of Sulfur Dioxide Gas in Reaction Kinetics [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Toxic / respiratory irritant [1]; individuals with asthma or respiratory conditions [1].	[2]
(b)	Aqueous sodium carbonate (Na ₂ CO ₃) or sodium hydrogencarbonate (NaHCO ₃) [1]; basic carbonate neutralizes acidic SO ₂ gas: $\text{SO}_2(\text{g}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{SO}_3^{2-}(\text{aq}) + \text{CO}_2(\text{g})$, trapping it as harmless aqueous sulfite [2].	[3]
(c)	Work in a well-ventilated laboratory with windows open or in an operating fume cupboard [2].	[2]
(d)	$\text{SO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SO}_3 + \text{H}_2\text{O}$ [3] (or $\text{SO}_2 + \text{NaOH} \rightarrow \text{NaHSO}_3$).	[3]

Question 10: Homogeneous Catalysis of the Peroxodisulfate–Iodide Reaction by Transition Metal Ions [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Both S ₂ O ₈ ²⁻ and I ⁻ are negatively charged anions; strong electrostatic repulsion between like charges creates a high activation energy barrier [2].	[2]

(b)	Step 1: $2\text{Fe}^{3+}(\text{aq}) + 2\text{I}^{-}(\text{aq}) \rightarrow 2\text{Fe}^{2+}(\text{aq}) + \text{I}_2(\text{aq})$ [2]; Step 2: $2\text{Fe}^{2+}(\text{aq}) + \text{S}_2\text{O}_8^{2-}(\text{aq}) \rightarrow 2\text{Fe}^{3+}(\text{aq}) + 2\text{SO}_4^{2-}(\text{aq})$ [2].	[4]
(c)	Each catalyzed step involves an oppositely charged cation and anion (Fe^{3+} with I^{-} , and Fe^{2+} with $\text{S}_2\text{O}_8^{2-}$), so attractive electrostatic forces facilitate collision [2].	[2]
(d)	Fe^{2+} can be oxidized by $\text{S}_2\text{O}_8^{2-}$ first to form Fe^{3+} , which then oxidizes I^{-} ; the cycle is closed and catalyst is regenerated in both cases [2].	[2]